# Provider and Patient Attitudes Regarding Sexual Health in Young Women With Cystic Fibrosis

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abstract

**OBJECTIVE:** To explore the attitudes, preferences, and experiences of patients with cystic fibrosis (CF) and CF providers toward sexual and reproductive health (SRH) care for young women with CF.

**METHODS:** Young women with CF aged 18 to 30 years from a US CF care center and pediatric and adult CF program directors from a national sample participated in qualitative interviews investigating their experiences regarding SRH care and their attitudes and preferences toward SRH care provision in the CF setting. Interviews were audio-recorded, transcribed, and coded by using a thematic analysis approach.

**RESULTS:** Twenty-two patient participants and 16 CF program directors were interviewed. Themes shared by both groups included the importance of SRH discussion in the CF care setting, patient and provider discomfort as a barrier to SRH care, and the need for SRH educational resources and provider training to improve SRH care. Providers highlighted the lack of standardization around SRH care in the current CF care model. Patients desired SRH educational resources coupled with early SRH discussions initiated by their CF provider.

**CONCLUSIONS:** Both CF providers and patients agree that the CF provider has a fundamental role in providing CF-specific SRH care. Educational resources coupled with individualized SRH discussions may facilitate improved SRH care for young women with CF. Investigation into the implementation of SRH education and services into pediatric-onset chronic disease care models is needed.

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Dr Kazmerski conceptualized and designed the study, conducted the patient and provider interviews, drafted the initial manuscript, and revised the manuscript; Drs Borrero, Weiner, and Tuchman aided in the design of interview scripts and critically revised the manuscript; Drs Pilewski and Orenstein reviewed and revised the data collection materials and manuscript; Dr Miller provided mentorship to Dr Kazmerski throughout the project, aided in the design of interview scripts, and critically revised the manuscript; and all authors approved the final manuscript as submitted.

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**WHAT'S KNOWN ON THIS SUBJECT:** Young women with cystic fibrosis (CF) face unique disease-specific sexual and reproductive health (SRH) concerns. Patients prefer to receive CF-specific SRH information from CF care providers, but major barriers to optimal SRH care exist in the current system.

**WHAT THIS STUDY ADDS:** This is the first study to assess the attitudes and experiences of both young women with CF and CF care providers toward SRH and identify facilitators to effective SRH care for this population.

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**TABLE 1** Key Interview Questions

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|---|
| Patient interviews |
| Has anyone on your CF care team ever discussed…(puberty, urinary incontinence, vulvovaginal candidiasis, sexual activity, contraception, fertility/pregnancy)…with you? |
| Would you like to talk about any of these SRH issues with your CF doctor? |
| Ideally, how would you like SRH care to occur? |
| How would you want to hear about SRH information specific to CF? |
| What would you like to tell other girls or women about your experiences in discussing SRH with your CF care team? |
| What would you like to tell CF providers about your experiences in discussing SRH with your CF care team? |
| Do you have any advice for your CF care team on how to make discussions about SRH issues more relevant or comfortable? |
| In your opinion, how can your CF doctor better support women such as yourself with SRH concerns? |
| Provider interviews |
| Do you think there is a role for the CF physician or team in discussing female patients' SRH? |
| Who do you feel should be responsible for managing/discussing SRH issues in your female patients with CF? |
| How are SRH issues/discussions currently addressed by you or your CF team (including puberty, urinary incontinence, vulvovaginal candidiasis, sexual activity, contraception, and fertility/pregnancy)? |
| Ideally, how would reproductive and sexual issues be addressed in your female patients with CF? |
| Tell me about the barriers that might inhibit talking about SRH issues with your female patients with CF. |
| As a program director, do you feel anything would be helpful in facilitating addressing SRH issues in your female CF population? |

Adolescent girls with chronic conditions are faced with both general and disease-specific sexual and reproductive health (SRH) concerns. Young women 15 to 24 years old experience the highest rates of sexually transmitted infections, unintended pregnancy, and intimate partner violence.[1–3] In addition, young women with chronic disease have SRH complications unique to their conditions.[4]

In particular, young women with cystic fibrosis (CF) face important disease-specific SRH concerns, including pubertal and menstrual delay, increased risk of vulvovaginal candidiasis, higher incidence of urinary incontinence, problems with sexual functioning, concerns regarding contraceptive choice, potentially decreased fertility, and adverse effects of pregnancy.[5–16] As the median life expectancy for patients with CF is now in the fourth decade of life,[17] an increasing number of patients require access to preventive services and care relevant to adolescence and the transition to adulthood, including SRH.

Limited literature exists regarding patient experiences and preferences toward SRH care in young women with CF.[18] Available data suggest that women with CF lack important SRH information critical to health promotion and quality of life[6,19,20] and feel the timing and content of SRH discussions in current CF care are inadequate.[21] In addition to patient dissatisfaction, suboptimal SRH care may result in serious health implications, such as exposure to teratogenic drugs, unintended pregnancies, or hypercoagulability risks.[9,22,23] Previous literature also has suggested that patients prefer to receive CF-specific SRH information from their CF care provider.[24]

To our knowledge, no studies have examined both female patient and subspecialty provider perspectives on SRH care in a pediatric chronic disease model. In this study, we conducted in-depth interviews with young women with CF and CF providers to identify attitudes, experiences, and barriers/facilitators to SRH care. Elucidating attitudes and preferences of these 2 stakeholder groups around SRH in the CF care model can guide interventions that improve care provision for this population and perhaps other pediatric-onset chronic disease states.

## METHODS

### Young Women With CF

We recruited a sample of women with CF ages 18 to 30 years from an accredited adult US CF care center during inpatient or outpatient visits between June and December 2014. Twenty-two participants completed semistructured, individual in-person interviews investigating SRH care experiences and attitudes toward SRH care in the CF setting. The principal investigator (PI) or a research assistant conducted the interviews in a private space after collecting participant demographic information, including CF and SRH history via a brief survey. The interviewers were not directly involved in the participants' clinical care. Interviews were structured by key questions and probes intended to guide conversation (Table 1). All interviews were audio-recorded and transcribed verbatim. Thematic saturation was reached after the 13th interview, suggesting that this sample was adequate for capturing the range of responses; additional interviews were conducted to ensure the sample was balanced by age and that no new themes emerged.[25] The study was approved by the University of Pittsburgh Institutional Review Board (PRO14030382).

### CF Providers

The Cystic Fibrosis Foundation (CFF) accredits a network of care centers that are usually composed of separate pediatric and adult



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**TABLE 2** Patient and Provider Characteristics

| | |
|---|---|
| Patient characteristic | |
| Age, y, mean (SD) | 25.1 (2.81) |
| Self-rating of CF severity, % (*n*) | |
| Mild | 41 (9) |
| Moderate | 46 (10) |
| Severe | 14 (3) |
| Age of menarche, y, mean (SD) | 13.6 (2.63) |
| Age of first sexual intercourse with male partner, y, mean (SD) | 17.9 (2.8) |
| Provider characteristic | |
| Age, y, mean (SD) | 55.4 (7.3) |
| Male, % (*n*) | 75 (12) |
| Program type, % (*n*) | |
| Adult | 31 (5) |
| Pediatric | 56 (9) |
| Combined | 13 (2) |
| Years post fellowship training (>15), % (*n*) | 88 (14) |
| Center size, % (*n*) | |
| >450 patients | 44 (7) |
| 250–450 patients | 31 (5) |
| <250 patients | 25 (4) |

programs. The directors of these programs are responsible for ensuring care is provided according to CFF guidelines. We focused our interviews on CF program directors (rather than all CF providers) because of their expertise in CF care and ability to reflect on overall center practices in addition to personal practice. As sites may be affected by regional differences in SRH education and attitudes, we selected program directors from geographically diverse institutions. Thirty potential participants were selected based on geographic diversity and contacted via E-mail by the CFF and PI. Sixteen directors agreed to be interviewed (a 53% participation rate) between April and August 2014.

The PI conducted semistructured, individual phone interviews, and explored attitudes toward SRH care, timing and content of SRH discussions, and potential barriers and facilitators to female CF SRH care (Table 1). Demographic information assessing participants' training, experience, and program characteristics was collected. Interviews were audio-recorded and transcribed verbatim. Thematic saturation in this homogeneous sample was reached after the eighth interview.[25]

## Data Analysis

Interview transcripts were analyzed through an iterative process of coding to identify themes. The PI developed 1 initial set of codes for patient participants and 1 for providers. The PI and research assistant then independently reviewed each transcript to apply the appropriate initial codebook and identify additional codes. By using a consensus coding approach, after the first few transcripts were independently coded, the coders reviewed their coding, discussed any discrepancies, and defined any new codes; this was iteratively repeated with subsequent transcripts until no new codes emerged. A senior co-investigator (EM) adjudicated any differences in interpretation. The final coding scheme was applied to all transcripts and central themes were identified. Representative quotations were selected from the transcripts to illustrate themes identified. ATLAS.ti 5.0 (Scientific Software Development GmbH, Berlin, Germany) was used to facilitate data management and coding.

## RESULTS

The patient sample included 22 participants, with a mean age of 25.1 ± 2.81 years. The provider sample included 16 program directors (9 pediatric, 5 adult, and 2 combined). Additional participant characteristics are shown in Table 2.

Several major themes and subthemes emerged during the interviews (Table 3). Themes shared by both patients and providers included the importance of SRH discussion in the CF setting, patient and provider discomfort as a barrier to SRH care, and the need for educational resources and provider training to improve SRH care. Patients were clear in their desire for SRH educational resources coupled with standardized provider discussions. They also preferred early SRH discussions initiated by the CF team. The shared themes are presented first, followed by additional themes raised by patients only. Additional illustrative quotes are included in Table 3.

### Theme 1: SRH Is Important to Discuss in the CF Care Setting

Patient participants believed SRH discussions were important and CF providers have a key role in SRH care. One woman stated, "It's always just like you don't talk about it [SRH], it's one of those things that's left to the side, it's [...] like they [CF providers] feel it's not as important as everything else, but sometimes it is. I mean, it [SRH] wasn't life or death threatening, but it could've changed my life a lot." A subtheme that emerged was the function of the CF provider as a de facto primary care provider (PCP). "I mean, I have a PCP, but I don't see them often. I usually go to the CF team for everything," 1 participant expressed.

Provider participants agreed SRH issues were highly important to discuss. One participant stated, "We've been so focused on nutrition and liver disease and lung disease and diabetes, but now that [...] quality of life continues to improve,



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**TABLE 3** Key Themes for SRH Care in CF Identified by Patients and Providers

| Theme | Representative Quotation(s) | |
|---|---|---|
| | Patients | Providers |
| 1. SRH is important to discuss in the CF care setting | "It's always just like you don't talk about it [SRH], it's one of those things that's left to the side, it's […] like they [CF providers] feel it's not as important as everything else, but sometimes it is. I mean, it [SRH] wasn't life or death threatening, but it could've changed my life a lot." | "We've been so focused on nutrition and liver disease and lung disease and diabetes, but now that […] quality of life continues to improve, this will be a big issue, a more important issue for everyone." |
| 2. Patient and provider discomfort around SRH is a major barrier to care | • Patient discomfort with SRH discussion | |
| | "Sometimes women are afraid to speak up and they keep these things [SRH issues] personal to them… and might feel uncomfortable." | "I think the number 1 [reason] is that a lot of the younger women are …embarrassed, especially because I'm a middle aged man, they're just a little embarrassed to bring it up." |
| | • Provider discomfort with SRH discussion | |
| | "I don't think they [CF providers] think about vaginas. I just think they like to think about lungs." | "You do what you're comfortable with. I'm not good at fielding questions about sexuality, so I probably don't bring it up as often as I should." |
| 3. Educational resources coupled with standardized provider discussions would facilitate SRH care | • Patient/family SRH educational resources | |
| | "Sometimes, if people were to feel uncomfortable… maybe be given a pamphlet. Or some papers that have Web sites that you can, you know, go on… or maybe there would be an online thing where you can actually ask questions, kinda like be anonymous because maybe some people are embarrassed." | "…a concise booklet that was […] very accurate [with] all the different [SRH] subjects at a comprehensive level. Because some [patients] probably don't necessarily want to talk about it in the clinic, but … we could provide accurate information to them that they could access at their own convenience." |
| | • Standardized provider discussions | |
| | "It's [SRH discussion] important. And, sometimes, [the CF provider] may have to be the initiator in these kinds of, you know, issues. Sometimes women are afraid to speak up and keep these things personal to them…and they [patients] might feel uncomfortable. The doctor should say, 'Hey, is there something – anything we could discuss about, you know, sexual development or things like that or pregnancy?'" | "[T]here's definitely not a systematic approach. I think for us it's very provider-dependent. It seems for our adult and adolescent providers, it's part of what they've been trained to do…I think our pediatricians aren't quite as systematic with it or as astute with how to approach it." |
| | "Learn to get comfortable with it. Because, CF affects every part of your health: it affects everything. I mean we talk about our lungs and our bowels. [We] might as well talk about the other stuff [SRH], too. Get comfortable with it!" | "I think one way [to improve SRH care provision] is to have clear transition topics that are brought up on a regular basis that need to be addressed. So that it [SRH] becomes a more routine part of annual visits and potentially even quarterly visits; it's just part of the issues that need to be addressed with regularity. So, regularity becomes familiarity." |
| | • Individualized SRH care provision | |
| | "I think it depends on the kid…because everybody is so different! There're some kids that are not very open and would rather just read it [SRH information] and there's some kids that are like, 'Hey…I just want somebody to straight up talk to me about this!' So... I think there should be options." | |



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**TABLE 3** Continued

| Theme | Representative Quotation(s) | |
|---|---|---|
| | Patients | Providers |
| 4. Women with CF prefer early, open-minded SRH discussions initiated by the CF team | • SRH discussions should occur at first sign of puberty or earlier<br>"[SRH] was brought up in school when I was in 4th or 5th grade, so I was probably 9…I think between 8 and 10, depending on if puberty is starting, I think you should be informed."<br>"Honestly, for me, the easiest thing would be to just start [SRH discussions] young and have it be an expectation. We walk in here and know that people are going to talk to us about bowel movements, that's just part of what we know is gonna be asked. So, if you start [SRH discussions] at a young age, I think it just becomes part of the routine and it doesn't become as uncomfortable as it would be."<br>• Preference for a stepwise approach to SRH discussions<br>"I think that they should start talking about something to do with it [SRH] at a younger age. So then when you grow, and hopefully continue to see the same doctor as you grow up, you feel comfortable with that doctor and discussing the rest of the topics."<br>• CF team should initiate SRH discussions<br>"I just think it would be good for the doctors to bring [SRH] up more because…when I was younger, I never even thought to say anything about it. And then, by the time I was old enough, it was well past the age…of needing concern." | |

this will be a big issue, a more important issue for everyone." When asked who should be responsible for discussing and managing SRH in female patients, all program directors stated that CF care providers have a fundamental role. Many program directors echoed the subtheme that, despite encouraging patients to have separate primary care, the CF care team often functions as a PCP and, therefore, the discussion of SRH (especially disease-specific issues) is primarily the CF center's responsibility.

### Theme 2: Patient and Provider Discomfort Around SRH Is a Major Barrier to Care

Both groups identified patient and provider discomfort as a major barrier to effective SRH discussion and care in young women with CF. One woman said: "Sometimes women are afraid to speak up and they keep these things [SRH] personal to them…and might feel uncomfortable." Another stated, "I don't think they [CF providers] think about vaginas. I just think they like to think about lungs."

When discussing patient discomfort, 1 provider stated, "I think the number 1 [reason] is that a lot of the younger women are … embarrassed, especially because I'm a middle-aged man, they're just a little embarrassed to bring it up." Provider discomfort was described in this way: "You do what you're comfortable with. I'm not good at fielding questions about sexuality, so I probably don't bring it up as often as I should."

### Theme 3: Educational Resources Coupled With Standardized Provider Discussions Would Facilitate SRH Care

Patient participants explicitly recommended in-person discussion with their CF provider plus provision of appropriate SRH education resources. Many respondents stated the need for individualized SRH care: "I think it depends on the kid… because everybody is so different! There're some kids that are not very open and would rather just read it [SRH information] and there's some kids that are like, 'Hey…I just want somebody to straight up talk to me about this!' So... I think there should be options." Many women felt that some sit-down conversation with the CF provider would be vital to "get any confusion [about SRH] out of the way." No preference between written or online educational materials emerged. Most patients encouraged other women with CF to become comfortable with SRH discussion in the CF care setting, "…we talk about our lungs and our bowels. [We] might as well talk about the other stuff [SRH], too. Get comfortable with it!"

Almost all program directors felt that age-appropriate patient SRH education resources would be helpful for young women with CF and their families. One director described the ideal resource as "a concise booklet that was […] very accurate [with] all the different [SRH] subjects at a comprehensive level. Because some



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[patients] probably don't necessarily want to talk about it in the clinic, but …we could provide accurate information to them that they could access at their own convenience."

Program directors reported a lack of a "standardized approach" and, thus, wide variation in SRH discussions based on care provider. Among provider practices, content of SRH discussions varied greatly, with some providers reporting routinely taking full SRH histories and others relying on patients or parents to bring up concerns. Timing of initial SRH discussions among pediatric program directors ranged from menarche to late adolescence. One provider commented:

> [T]here's definitely not a systematic approach. I think for us it's very provider-dependent. It seems for our adult and adolescent providers, it's part of what they've been trained to do…I think our pediatricians aren't quite as systematic with it or as astute with how to approach it.

Many providers felt that creation of "formalized" assessments or "a routine approach" directed by universally accepted guidelines would be helpful. One participant stated, "…one way [to improve SRH care provision] is to have clear transition topics that are brought up on a regular basis…so that it [SRH] becomes a more routine part of…issues that need to be addressed with regularity. So, regularity becomes familiarity."

Regarding SRH discussions with patients, most program directors felt that education and training programs for CF providers in SRH topics would be helpful. Many commented that using international conferences as a forum to discuss SRH research and offer training to CF providers would be an opportunity to improve provider acceptance of these issues. Some participants also suggested that SRH care could be improved by inclusion of SRH outcomes in the US CF registry and further research around female CF SRH topics.

### Theme 4: Women With CF Prefer Early, Open-Minded SRH Discussions Initiated by the CF Team

Patient participants felt SRH discussions should start with the first signs of expected puberty or before. As 1 woman stated, "[SRH] was brought up in school when I was in fourth or fifth grade, so I was probably 9…I think between 8 and 10, depending on if puberty is starting, I think you should be informed."

Many patient participants felt a "stepwise approach" to SRH discussions introducing pertinent topics in a gradual fashion would be helpful in limiting patient discomfort. As 1 participant said, "I think that they should start talking about something to do with it [SRH] at a younger age. So then, when you grow and hopefully continue to see the same doctor as you grow up, you feel comfortable with that doctor and discussing the rest of the topics." However, some women disagreed and felt that it was important to discuss all the SRH effects of CF early to assure young patients that these differences from their peers are "normal" and to allow them to make informed SRH decisions.

Nearly all women interviewed believed that the CF team should initiate the conversation, as many remarked that younger patients would not volunteer this information. One participant stated, "I just think it would be good for the doctors to bring [SRH] up more because…when I was younger, I never even thought to say anything about it. And then, by the time I was old enough, it was well past the age…of needing concern."

## DISCUSSION

This is the first study to assess the attitudes and experiences of both patients and providers toward SRH care among young women with CF. In this sample of program directors and young women with CF, there was agreement that SRH care for young women with CF is important, CF providers have a fundamental role in SRH care provision, and systematic approaches are needed.

Most participants saw CF providers as having a key role in SRH care and often functioning as de facto PCPs. This subtheme reflects previous research demonstrating that adolescents with CF or sickle cell disease commonly report their subspecialist as their main physician.[26] The use of the subspecialty clinic as a medical home/PCP and the impact on quality of care should be further investigated both in CF and other pediatric-onset chronic diseases.

Previously, we described obstacles to the provision of SRH care in the CF setting as identified by CF program directors only.[27] With this study, we asked both patients and providers to reflect on not only barriers, but also factors that may facilitate improved SRH care for young women with CF. Additionally, the lack of SRH training for CF providers may impede such care in the current multidisciplinary CF care model.[28]

Frayman and Sawyer[5] recently presented a framework of SRH care provision and education for patients with CF and their families. In this model, the primary CF provider is responsible for ensuring that SRH issues are addressed. In the first decade of life, parents are educated and discuss SRH topics with their children; at the start of puberty, the focus then shifts to routine confidential discussions between providers and patients. The model also highlights communication and coordination of care among the CF team, PCPs, and specialist services, such as adolescent medicine and obstetrics/gynecology.

The role of the CF team in SRH care posited in this model is congruent with our findings that young women with CF and CF providers value SRH care and the role of the CF



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team in its delivery. Although the suggested framework offers a general organizational structure for SRH care in the CF setting, there is currently no specific approach to implementation or tools to facilitate SRH care in the population of young women with CF. Further investigation of stakeholder attitudes (including adolescent patients, parents/guardians, CF providers, and adolescent or women's health specialists) toward CF SRH care delivery can help determine the best timing and approach for offering SRH care and education. Additionally, greater attention to defining relevant SRH patient-centered outcomes is necessary.

In this qualitative study, patient and provider groups agreed that educational resources coupled with standardized provider discussions would facilitate better SRH care. Unfortunately, educational resources for young women with CF highlighting disease-specific components of SRH are currently lacking. As effective educational tools exist in other chronic disease states with disease-specific SRH concerns (such as diabetes),[29–32] adapting these resources to the CF-specific SRH needs and patient-centered outcomes of young women with CF and their families is feasible and an important next step. Improvement in routine provider-initiated SRH discussions can be facilitated in the creation of SRH training programs for the CF specialist. Again, adapting existing SRH training vehicles[33–35] to the CF provider population through stakeholder engagement may be warranted.

Similar disease-specific concerns regarding SRH care are present in other settings, such as transplant medicine and oncology as well as in conditions such as pulmonary hypertension, sickle cell disease, and rheumatologic disease. As serious and even mortal SRH implications exist in these disease states,[36–38] analogous investigation into SRH care is needed. The findings of this study may help inform other subspecialty providers of both a patient need and an avenue to improve existing comprehensive care models.

This study has several important limitations. First, qualitative research is used to explore participant perspectives in great depth. Sample sizes are generally determined by thematic saturation and generalizability is limited. These narratives are critical to understanding the complex factors that influence SRH care. In addition, our study used a purposive sample of US CF program directors. Thus, findings cannot be generalized to all CF care providers (ie, those who are not directors). The gender and discipline (pediatric versus adult) of providers may also affect SRH attitudes. Also, results from patient interviews may not be applicable to all young women with CF, as the patient interviews were conducted at a single center.

Quantitative investigation around SRH concerns, care utilization, and patient preferences related to SRH care among young women with CF is ongoing. Similar investigation of a broader array of CF provider attitudes and practices around SRH care (including the influence of gender and provider discipline) is needed. Given the preferences for SRH care elucidated in this work, provider training tools and educational resources relevant to important SRH patient-centered outcomes should be developed. Furthermore, many patient participants indicated that they would prefer to have SRH discussions with a nurse or doctor on the CF care team; future work should identify the optimal person to deliver SRH information from patient and provider perspectives. Future SRH care interventions should be tested for feasibility, acceptability, and effectiveness in the population of young women with CF. National guidelines for high-quality SRH care provision in CF clinical care may be developed by using this evidence.

## CONCLUSIONS

Providers and patients agree that the CF team has a fundamental role in SRH care. Educational resources coupled with individualized SRH discussions may facilitate improved SRH care for young women with CF. Investigation around implementation of SRH education and services into the CF care model is needed. In the future, this may be used as a guide for comprehensive SRH care for young women with other pediatric-onset chronic diseases.

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### ABBREVIATIONS

CF: cystic fibrosis
CFF: Cystic Fibrosis Foundation
PCP: primary care provider
PI: principal investigator
SRH: sexual and reproductive health

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